TRACI™

01 / SMART SITE SAFETY

Hook Cam

A closer view of every lift.

One connected kit for lifting operations. Video, height sensing, 4G connectivity and local alerts — supported by a quick-swap LiFePO4 battery.

SEE THE LIFT.
UNDERSTAND THE MOMENT.



Concept kit visual. Final configuration may vary.

BUILT FOR LIFTING OPERATIONS

01
SEE & MEASURE

A closer view of the hook and load, with a distance reference for clearance.

02
CONNECT & ALERT

4G connectivity links site teams. A local siren provides audible alerts.

03
POWER THE SHIFT

Quick-swap battery power supports flexible deployment at the hook.

COMPLETE SYSTEM CONFIGURATION

06 COMPONENTS

01
reCamera Pro 4GB

4K edge AI camera



02
4G router

Cellular connectivity



03
Local siren

Audible site alerts



04
Quick-swap battery

24 V-class / 500-600 Wh target



05
PoE power module

Camera power + Ethernet



06
Benewake TF02-Pro

Height / distance reference



Illustrative configuration. Final component models and appearance are subject to confirmation.

SEAMLESS SITE INTEGRATION

Open 4S

Connected your way.

Bring data from existing site systems into TRACI, or share Embuilded-connected device data with third-party platforms in their preferred formats. Flexible two-way integration helps connect safety workflows without replacing existing infrastructure.

Embuilded™
Embedded intelligence for the built world.

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02 / TECHNICAL SPECIFICATIONS

Camera & height sensing

Core hardware for video capture and distance measurement.

reCamera Pro 4GB

SEEED STUDIO



4K / 30 fps

3 TOPS EDGE AI

Image sensor	SC850SL; up to 4K at 30 fps
Processor	RV1126B; quad-core Cortex-A53, 1.2 GHz
AI accelerator	3 TOPS; INT8 / INT16 mixed computation
Memory / storage	4 GB LPDDR4; 16 GB eMMC
Expandable storage	microSD card slot, up to 512 GB
Connectivity	Gigabit Ethernet; Wi-Fi; Bluetooth 5.2
Power input	PoE or 12-24 V DC via 5525 barrel jack
Other interfaces	USB Type-C OTG; 2 x CAN; 2 x GPIO

Benewake TF02-Pro

SINGLE-POINT ToF LiDAR



0.1-40 m

AT 90% REFLECTIVITY

IP65 / 50 g

Measurement range	0.1-40 m at 90% reflectivity; 0.1-13.5 m at 10% reflectivity
Accuracy	±5 cm (0.1-5 m); ±1% (5-40 m)
Resolution	1 cm
Frame rate	1-1000 Hz; 100 Hz default
Field of view	3° (theoretical)
Ambient light	Up to 100 klux
Data interface	UART / I²C; 3.3 V LVTTL
Supply / power	5-12 V DC; ≤1 W
Environment	IP65; -20 to +60 °C operating
Size / weight	69 × 41.5 × 26 mm; 50 g with cable

MEASUREMENT CONDITIONS

Sensor range depends on target reflectivity and viewing conditions. Manufacturer accuracy is specified on a 90% reflective target at 25 °C. Height reference depends on sensor alignment and mounting position.

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Battery & supporting equipment

Portable energy, cellular communication and local audible alerting.

QUICK-SWAP BATTERY

TARGET SPECIFICATION



500-600 Wh

24 V-CLASS LiFePO4

Reference pack photo only. Final quick-swap housing to be confirmed.

Chemistry	Lithium iron phosphate (LiFePO4)
System voltage	24 V-class; final nominal voltage TBC
Energy capacity	500-600 Wh design target
Battery exchange	Quick-swap pack; connector / latch TBC
Protection	BMS and fused output required
Charging	Matched LiFePO4 charger; rating TBC
Runtime	Dependent on usable energy and total load
Size / mass	Subject to final pack and enclosure design

REGULATED POWER DISTRIBUTION

Use a regulated DC / PoE supply matched to the battery voltage range. reCamera Pro accepts 12-24 V DC; TF02-Pro requires 5-12 V DC. A 24 V-class battery must not be assumed compatible with direct connection.

SUPPORTING EQUIPMENT

MODELS TO BE CONFIRMED

COMPONENT	CONFIGURATION REQUIREMENTS
 **4G router** Model TBC	4G LTE with deployment-region bands; Ethernet uplink; external antennas. Specify VPN / remote access, DC input range and power consumption.
 **PoE module** Model TBC	DC-input PoE injector with regulated output. Confirm camera PoE standard / power class, battery voltage coverage and Ethernet speed.
 **Local siren** Model TBC	External audible alert unit with a matched DC supply and control interface. Specify dB(A) at 1 m, tone, duty cycle and ingress protection.

Battery, router, siren and PoE values are configuration targets. Final system dimensions, weight, enclosure rating and operating time depend on the selected assembly.

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